

DATA REQUIERMENT DOC (DRD)

Project Title: E-Commerce Retail Database System ("ShopSphere")
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1. DOCUMENT PURPOSE

The purpose of this Data Requirement Document is to translate the business requirements defined in the Requirement Analysis Document (RAD) into technical data specifications. This document defines the logical data model, entity structures, attribute data types, relational constraints, data security boundaries, and reporting data extraction rules required to implement the ShopSphere RDBMS.

2. LOGICAL DATA ENTITIES & ATTRIBUTE SPECIFICATIONS

Based on the four core business domains, the following entities and their attributes are required. Primary Keys (PK), Foreign Keys (FK), and Unique constraints are noted.

2.1 User & Account Management Domain

Stores customer profiles, login credentials, and system roles.

Attribute	Data Type	Constraints	Description / Rule
UserID	INT	PK, Auto-Increment	Unique system identifier for the user.
Name	VARCHAR(100)	NOT NULL	Full name of the user.
Email	VARCHAR(255)	UNIQUE, NOT NULL	Acts as login credential; cannot be

			shared.
PasswordHash	VARCHAR(255)	NOT NULL	Encrypted login password.
ContactNumber	VARCHAR(15)	NULL	User's phone number. (Blanked out if account Inactive).
RegistrationDate	TIMESTAMP	NOT NULL, Default CURRENT_TIMESTAMP	System-generated date of registration.
Role	ENUM('Customer', 'Manager', 'Staff', 'Admin')	NOT NULL	Defines actor profile and database access privileges.
AccountStatus	ENUM('Active', 'Inactive')	NOT NULL, Default 'Active'	Cannot be hard-deleted. Set to 'Inactive' upon deletion.

Stores multiple shipping addresses per user.

Attribute	Data Type	Constraints	Description / Rule
AddressID	INT	PK, Auto-Increment	Unique identifier for the address.
UserID	INT	FK (Users.UserID), NOT NULL	Links address to the owning user.
AddressLine	VARCHAR(255)	NOT NULL	Street address details.
City, State, ZipCode	VARCHAR/ varchars	NOT NULL	Location hierarchy.
IsDefault	BOOLEAN	NOT NULL, Default FALSE	Tags one address as the default shipping location.

2.2 Product Catalog & Inventory Domain

Organizes products into distinct groupings.

Attribute	Data Type	Constraints	Description / Rule
CategoryID	INT	PK, Auto-Increment	Unique identifier for the category.
CategoryName	VARCHAR(100)	UNIQUE, NOT NULL	E.g., Electronics, Apparel.

(Derived from Report 1 requirement)

Stores supplier restocking contact information.

Attribute	Data Type	Constraints	Description / Rule
SupplierID	INT	PK, Auto-Increment	Unique identifier for the supplier.
SupplierName	VARCHAR(150)	NOT NULL	Name of the supplier.
ContactInfo	VARCHAR(255)	NOT NULL	Email/Phone for restocking alerts.

Houses item details, pricing, and inventory stock levels.

Attribute	Data Type	Constraints	Description / Rule
SKU	VARCHAR(50)	PK	Stock Keeping Unit – unique product identifier.
ProductName	VARCHAR(255)	NOT NULL	Descriptive name of the item.
Price	DECIMAL(10,2)	NOT NULL, CHECK (Price > 0)	Current selling price; strictly positive.
StockQuantity	INT	NOT NULL, CHECK (StockQuantity >= 0)	Current available inventory.
CategoryID	INT	FK (Categories.CategoryID), NOT NULL	Every product must belong to at least one primary Category.
SupplierID	INT	FK (Suppliers.SupplierID)	Link to supplier for restocking.
AvailabilityStatus	ENUM('In Stock', 'Out of Stock', 'Discontinued')	NOT NULL	Automatically toggles to 'Out of Stock' if StockQuantity = 0.

(Derived from Report 3 & Constraints)

Captures customer ratings and reviews for analytical reporting.

Attribute	Data Type	Constraints	Description / Rule
ReviewID	INT	PK, Auto-Increment	Unique identifier for the review.
SKU	VARCHAR(50)	FK (Products.SKU), NOT NULL	Product being reviewed.
UserID	INT	FK (Users.UserID), NOT NULL	Customer writing the

			review.
Rating	INT	NOT NULL, CHECK (Rating >= 1 AND Rating <= 5)	Integer rating between 1 and 5 inclusive.
ReviewText	TEXT	NULL	Optional text feedback.
ReviewDate	TIMESTAMP	NOT NULL, Default CURRENT_TIMESTAMP	Date the review was posted.

2.3 Shopping Cart & Order Lifecycle Domain

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Temporary storage for items prior to checkout.

- **ShoppingCart**: CartID (PK), UserID (FK, UNIQUE - 1 cart per user), CreatedDate.
- **CartItems**: CartID (FK), SKU (FK), Quantity (INT, CHECK > 0).

The core transactional entity capturing the checkout event.

Attribute	Data Type	Constraints	Description / Rule
OrderID	INT	PK, Auto-Increment	Unique identifier for the order.
UserID	INT	FK (Users.UserID), NOT NULL	Linked to exactly one customer.
OrderDate	TIMESTAMP	NOT NULL	Time of checkout.
TotalAmount	DECIMAL(10,2)	NOT NULL, CHECK (TotalAmount > 0)	Total financial value of the order.
Status	ENUM('Pending', 'Payment Confirmed', 'Shipped', 'Delivered', 'Cancelled')	NOT NULL	Rigid lifecycle state machine.

Bridging table resolving the M:N relationship between Orders and Products.

Attribute	Data Type	Constraints	Description / Rule
OrderID	INT	FK (Orders.OrderID), PK (Composite)	Links to the parent order.
SKU	VARCHAR(50)	FK (Products.SKU), PK (Composite)	Links to the purchased product.

Quantity	INT	NOT NULL, CHECK (Quantity <= StockQuantity at checkout)	Number of items bought.
UnitPrice	DECIMAL(10,2)	NOT NULL, CHECK (UnitPrice > 0)	Historic Price Rule: Price at the exact second of purchase; immutable.

2.4 Payments & Shipping Operations Domain

1:1 relationship with Orders. Cannot progress without a valid OrderID.

Attribute	Data Type	Constraints	Description / Rule
PaymentID	INT	PK, Auto-Increment	Unique payment transaction ID.
OrderID	INT	FK (Orders.OrderID), UNIQUE	Enforces 1:1 relationship with Order.
PaymentMethod	VARCHAR(50)	NOT NULL	E.g., Credit Card, PayPal.
Amount	DECIMAL(10,2)	NOT NULL	Currency validated amount.
PaymentStatus	ENUM('Successful', 'Failed', 'Pending')	NOT NULL	Must be 'Successful' for Order to progress to 'Shipped'.

Logistics and tracking data, generated once order is shipped.

Attribute	Data Type	Constraints	Description / Rule
ShipmentID	INT	PK, Auto-Increment	Internal shipping record ID.
OrderID	INT	FK (Orders.OrderID), UNIQUE	Links to the order.
TrackingNumber	VARCHAR(100)	UNIQUE , NOT NULL	Carrier-generated unique tracking number.
CarrierName	VARCHAR(100)	NOT NULL	Shipping carrier (e.g., FedEx, UPS).
ShipDate	TIMESTAMP	NOT NULL	Date the item was dispatched.
DeliveryDate	TIMESTAMP	NULL	Populated only when carrier API confirms delivery (triggers Order Status -> Delivered).

(Derived from Admin constraints)
Tracks all System Admin transactions to prevent internal fraud.

- LogID (PK), AdminUserID (FK), ActionType (ENUM CREATE/UPDATE/DELETE), AffectedTable, Timestamp.

3. RELATIONAL MAPPING SUMMARY

Relationship Type	Entities Involved	Implementation Rule
1:N (One-to-Many)	Users to Addresses	One user can save multiple shipping addresses.
1:N (One-to-Many)	Categories to Products	One category contains multiple products.
1:N (One-to-Many)	Users to Orders	One customer can place multiple orders over time.
M:N (Many-to-Many)	Orders to Products	Resolved via OrderItems bridging table with composite PK.
1:1 (One-to-One)	Orders to Payments	OrderID is a UNIQUE FK in the Payments table.
1:1 (One-to-One)	Orders to Shipments	OrderID is a UNIQUE FK in the Shipments table.

4. DATA ACCESS CONTROL MATRIX (SECURITY)

Data privileges mapped to Actor Profiles to maintain system security boundaries.

Data Entity	Customer (Shopper)	Store Manager	Fulfillment Staff	System Admin
Users / Addresses	Read/Write (Own only)	Read (Aggregated only)	Read (Shipping Addr only)	Full CRUD (Logged in Audit)
Products / Inventory	Read only	Read/Write (Prices, Stock, Status)	Read (Packing details only)	Full CRUD (Logged in Audit)
Orders / OrderItems	Read/Write (Own cart/ orders)	Read (Sales stats only)	Read/Write (Status to Shipped)	Full CRUD (Logged in Audit)
Payments	Read (Own	NO ACCESS	NO ACCESS	Full CRUD

	payment status)	(Cannot view CC data)		(Logged in Audit)
Shipments	Read (Own tracking info)	Read (Logistics stats)	Read/Write (Input Tracking Num)	Full CRUD (Logged in Audit)
AuditLogs	NO ACCESS	NO ACCESS	NO ACCESS	Read Only (Immutable)

5. ANALYTICAL DATA EXTRACTION RULES (REPORTING)

The database must support the following aggregate SQL extraction logic:

Report 1: Low-Inventory Restock Alerts

- **Filter:** Products.StockQuantity <= 10 AND Products.AvailabilityStatus != 'Discontinued'
- **Join:** Products INNER JOIN Categories (on CategoryID) INNER JOIN Suppliers (on SupplierID)
- **Output Columns:** ProductName, SKU, CategoryName, StockQuantity, Suppliers.ContactInfo

Report 2: VIP Customer Loyalty Report

- **Filter:** Orders.OrderDate within the last 12 months AND Orders.Status = 'Payment Confirmed' / 'Shipped' / 'Delivered' (Successful states).
- **Join:** Users INNER JOIN Orders (on UserID)
- **Aggregation:** SUM(Orders.TotalAmount) grouped by UserID
- **Sort:** Descending by Total Amount
- **Limit:** Top 50 Rows
- **Output Columns:** Users.Name, Users.Email, SUM(TotalAmount) AS TotalSpent

Report 3: Category Sales Performance Directory

- **Filter:** Orders.OrderDate falling within a specified calendar month parameter.
- **Join:** Categories -> Products -> OrderItems -> Orders -> ProductReviews
- **Aggregation:** SUM(OrderItems.UnitPrice * OrderItems.Quantity) AS Total Revenue, SUM(OrderItems.Quantity) AS Items Sold, AVG(ProductReviews.Rating) AS AvgRating
- **Group By:** Categories.CategoryID

- **Output Columns:** CategoryName, TotalRevenue, ItemsSold, AvgRating
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6. NON-FUNCTIONAL DATA CONSTRAINTS & TRANSACTIONS

6.1 Strict Data Integrity Constraints

1. **Value Bounds:** CHECK (ProductReviews.Rating >= 1 AND ProductReviews.Rating <= 5)
2. **Positive Financials:** CHECK (Products.Price > 0 AND Orders.TotalAmount > 0 AND OrderItems.UnitPrice > 0)
3. **Stock Validation:** Checkout application logic must enforce CHECK (OrderItems.Quantity <= Products.StockQuantity) at the time of insertion.

6.2 Concurrency Controls (ACID Implementation)

- **Isolation (Row-Level Locking):** During simultaneous checkout for limited inventory, the RDBMS must execute SELECT StockQuantity FROM Products WHERE SKU = ? FOR UPDATE;.
- **Process:** The first transaction locks the row, processes payment, updates StockQuantity, and commits (releasing the lock). The second transaction then reads the updated row (Stock = 0) and gracefully fails, preventing overselling.
- **Atomicity (Rollback Protocol):** If payment gateway verification fails during checkout, the entire transaction (Order creation, OrderItems insertion, Inventory deduction) must be rolled back to preserve a consistent state.

6.3 Historical Auditing & Safe Deletion (Soft Deletes)

- **No Hard Deletion:** DELETE commands are permanently disabled for the Users table for all roles including System Admin.
- **Privacy Compliance Mechanism:** Upon account deletion request, an UPDATE transaction sets Users.AccountStatus = 'Inactive', and blanks out PII fields (Email = 'anon_[UserID]@null', ContactNumber = NULL).
- **Financial Audit Preservation:** FK constraints linking Orders.UserID to Users.UserID must use ON NO ACTION to ensure past order receipts and historical sales reports remain perfectly balanced even if the user profile is anonymized.